

Muru Music Health: Returning harmony to people with dementia through music therapy

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[About Muru Music](#)



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About Muru Music Health

Muru Music is a software technology company that offers the capability to create personalized, seamlessly generated music experiences. What makes Muru Music's model unique is its overarching vision to power all digital music experiences and its ability to provide an end-to-end personalized experience. While it can be applied to verticals ranging from hospitality to fitness, Muru Music's decision to start with music therapy for people living with dementia stems from a personal connection to the

Google Cloud results

- Coordinates an international team using Google Workspace solutions such as Meet for online conferencing and Drive for collaboration
- Speeds up the completion of heavy data processing tasks using Dataflow and AI Platform
- Drives efficiency and quick, effective decision-making by empowering engineers to

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disease, as well as a core belief in the power of personalized music listening as therapy.

Industries: Technology

Location: Australia

(<https://cloud.google.com/ai-platform/>), [Cloud](#)

[Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>)

[Dataflow](https://cloud.google.com/dataflow/) (<https://cloud.google.com/dataflow/>),

[BigQuery](https://cloud.google.com/bigquery/) (<https://cloud.google.com/bigquery/>)

[Monitoring](#)

[AI Platform](https://cloud.google.com/ai-platform/) (<https://cloud.google.com/ai-platform/>)

[Cloud CDN](https://cloud.google.com/cdn/) (<https://cloud.google.com/cdn/>)),

[Kubernetes Engine](#)

[Dataflow](https://cloud.google.com/dataflow/) (<https://cloud.google.com/dataflow/>)),

[Google Workspace](#)

[Logging](https://cloud.google.com/logging/) (<https://cloud.google.com/logging/>)

[Monitoring](https://cloud.google.com/monitoring/) (<https://cloud.google.com/monitoring/>)

[Cloud Storage](https://cloud.google.com/storage/) (<https://cloud.google.com/storage/>)

<https://workspace.google.com/products/docs/>

[Kubernetes Engine](#)

<https://cloud.google.com/kubernetes-engine/>

<https://workspace.google.com/products/ai/>

[Google Workspace](#)

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experiment
with new
approaches

Google Meet

(<https://workspace.google.com/products/meet/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

Music can be powerful. Nicc Johnson has known this since he was a young DJ, starting his professional career at some of the biggest clubs in the world. “Whether it was in Ibiza or Amsterdam or Shanghai, there was always an incredible connection on the dance floor,” he shares. “I wanted to understand what was happening, so I studied musicology.”

Nicc came to see the healing properties of music and how they could be used in therapy. Today, he’s the CEO and founder of [Muru Music](https://www.murumusic.com/) (<https://www.murumusic.com/>), the first digital music platform for people living with dementia. “I’ve experienced Dementia firsthand and have seen the devastating effects it can have on a person and their family,” he shares. “I also know the incredible power of music and want to ensure that personalized music therapy is available to anyone, at any time, anywhere.”

With a team that spans a wide range of talent, including musicologists, data scientists, engineers, and DJs, Muru Music uses cutting-edge AI technology to analyze the “vibe” of music and create recommendations based on factors such as someone’s hobbies, interests, and where they grew up. Muru Music uses artificial intelligence to create personalized music experiences for each of its users, helping them cope with some of the most

challenging symptoms of dementia, including agitation, anxiety, and loneliness. By playing songs that its users would have heard in their past, Muru Music harnesses the power of music to help them relax, experience positive memories, or enjoy entertainment, all while stimulating the brain, to provide a better quality of life and to reduce the negative behavioral and psychological effects of dementia.

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—Nicc Johnson, CEO and
founder, Muru Music

After initial trials in Australia, Nicc and his team were confident that the platform really worked. But as Muru Music Health grew, adding music partners and serving more people, it started to need more IT resources to keep providing the same level of service. The existing infrastructure was beginning to buckle under the demand, with outages in the data pipelines and slow performance. If Muru Music was to continue growing, it needed a new solution that was reliable and scalable. To achieve this, it chose Google Cloud.

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Muru Music Health is not like other streaming platforms. When a profile is created for someone living with Dementia, you answer a number of questions about the person's past, such as the

language they speak, where they grew up, and which songs, if any, they enjoyed. The idea is to generate as accurate a picture as possible of the user's history and work out which particular musicians and songs they might have listened to in the past.

Once the user's music history has been predicted, they decide on the purpose of their playlist. This can vary depending on the user, or even on their mood at that particular time. For example, they might want to use music for relaxation, to improve their mood, for exercise, or simply for entertainment.

With a focus on older music that is often hard to find, Muru Music Health partners with a number of large music labels to access their libraries and find the right songs. Each song is analyzed for particular sonic qualities, and the company's AI platform assigns a "vibe" score. The platform then creates the playlist using songs from the right time period and with the right vibes. It also asks for feedback from the user or their caregiver, using a simple "like" or "dislike" button, so that it can adapt the music selection on the fly according to each person's preference.

Making the process this simple for the end user requires a huge amount of work on Muru Music's part. "We're working with raw audio and analyzing millions of data points a day," says Nicc. When the company first started, it used a number of cloud solutions to build its infrastructure but ran into problems with their data architecture. With only a small team of six spread out across the world, Nicc couldn't just throw resources at the problem.

“Things weren’t running properly or were taking too long,” he explains. “Data processing tasks were taking months to run. It just wasn’t scalable. We needed a new infrastructure solution.”

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Scaling on demand with Google Cloud

Prompted by recommendations from its own engineers, Muru Music chose to run its Health platform on [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>). The engineers were particularly excited by the scaling opportunities afforded by [Google Kubernetes Engine](#) ([/kubernetes-engine](#)) (GKE). By using containers instead of virtual machines, Muru Music can rapidly scale its compute power without having to worry about setting up and configuring new machines.

GKE handles the main infrastructure, allowing Muru Music to offload much of the management overhead to Google Cloud, while [Cloud Logging](#) ([/logging](#)) and [Cloud Monitoring](#) ([/monitoring](#)) give engineers a clear view into operations at all times. The data platform relies on [Cloud Storage](#) ([/storage](#)) and [BigQuery](#) ([/bigquery](#)) to hold and analyze the data. [AI Platform](#) ([/ai-platform](#)) provides Muru Music with new tools and ways of adding to the existing AI platform and drawing out new insights. Meanwhile, [Dataflow](#) ([/dataflow](#)) has been instrumental in speeding up the company's data pipelines and making them more reliable.

"With Cloud Dataflow and the other data-focused solutions, we were able to parallelize our tasks in a way that we couldn't do before," says Nicc. "It meant we could work faster and smarter, processing more data in less time."

Organizing a team across borders and time zones can be a challenge, especially during a time of change like an infrastructure migration, but the company used Google Workspace (<https://workspace.google.com/>) to make sure that all employees were fully connected and working together, no matter where they are. At least twice a week, Nicc liaises with his team on Google Meet (<https://workspace.google.com/products/meet/>) for a video call. Workers can share documents and files on Drive (<https://workspace.google.com/products/drive/>), and even work on them together with live editing in Docs (<https://workspace.google.com/products/docs/>) and Sheets (<https://workspace.google.com/products/sheets/>).

“With Google Workspace, we can continuously evolve,” says Nicc. “We sync up at the beginning of the week and at the end of the week. Meanwhile, we’re all working on the same documents and putting comments and updates in Drive so everyone can see how we’re doing. It works beautifully.”

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Music therapy in real time, at scale, and geared for growth

Since migrating to Google Cloud, Muru Music Health has been able to scale its service along with its ambitions. Perhaps the biggest change for the company has been the speed at which it can accomplish complex data tasks. That speed allows the company to constantly add new music to its library and respond to user feedback more rapidly.

“Music therapy through a platform like this will only ever work if it’s in real time,” says Nicc. “With our new AI platform, we can deliver real-time responses to user feedback and change our playlists on the fly. Google Cloud is essential to creating the highly personalized, seamless service that we provide.”

Another benefit of Google Cloud is that it allows Muru Music to access as much compute power as it needs, without having to make large up-front investments in hardware or manpower. Today, Muru Music handles around two petabytes of data per week and has reduced processing time dramatically. “It’s cost savings, but for us, it’s also time savings that are key,” says Nicc. “One prediction task in particular would have taken more than four months to complete on our previous infrastructure. With Google Cloud, we did it in just seven hours.”

From the perspective of company culture, these time savings have changed the way that Muru Music runs its business, giving engineers more time to properly experiment with new techniques or features instead of worrying about maintaining pipelines. “One good engineer can run a set of experiments, and within a week I can see the results,” says Nicc. “It allows me to make decisions very, very quickly. Our whole work culture has become much more efficient.”

With the migration to Google Cloud complete, Muru Music is currently looking ahead to working with more people living with dementia. Over the next year or so, as it moves into the United States and the United Kingdom, the company will expand its use of

Cloud CDN to deliver the same levels of high-speed service that its existing users in Australia enjoy.

Muru Music is also looking to bring its Health platform not just to people with dementia but to people in aged-care homes who might need extra support. “We’ve found that the positive effects are the same for users in aged-care homes as for our users with dementia, in terms of how music stimulates the brain,” says Nicc. “Our goal is to make sure that access to music therapy is here to help anyone who needs it.”